

1. Perceptron learning algorithm (supervised learning algorithm):

A. supervised learning: when training a model (neuron network), we will offer not only "inputs x_i " but also corresponding "output y (ground truth)" as pair training example [inputs x_i , output y (ground truth)] for the model to learn.

B. processes of perceptron learning algorithm: step1. randomly initialize the weights w_i and given pair training example [inputs x_i , output y (ground truth)]
step2. select one pair training example [inputs x_i] for perceptron to compute corresponding "predicted output P_y "
step3. match predicted output P_y & output y (ground truth): [step3 is only for single perceptron, not fit for neural network]
if "($p_y < 0$) != ($y = +1$)" = "($p_y = -1$) != ($y = +1$)"
 ,then " add each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i]), y = +1$ "

if "($p_y > 0$) != ($y = -1$)" = "($p_y = +1$) != ($y = -1$)"
 ,then " subtract each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i]), y = -1$ "
step4. repeat "step2 & step3" until predicted output $P_y =$ output y (ground truth).

2. Learning rate (η): an example of hyperparameter! which is not a parameter adjusted by perceptron learning algorithm but by user.

3. Decision boundary equation:

A. In two-input perceptron, there is a line ($w_0x_0 + w_1x_1 + w_2x_2 = 0$) in 2D graph, the line can divide the graph into 3 parts. $\Rightarrow$ Part1. on the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0$
Part2. one side of the line: $w_0x_0 + w_1x_1 + w_2x_2 > 0$
Part3. the other side of the line: $w_0x_0 + w_1x_1 + w_2x_2 < 0$

B. the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0 \rightarrow x_2 = -\frac{w_1}{w_2}x_1 - \frac{w_0}{w_2}$
slope of line
y-intercept of line

